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# Effect of different environmental conditions on the growth and development of Black Soldier Fly Larvae and its utilization in solid waste management and pollution mitigation

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ABSTRACT

The current waste treatment techniques in developing countries do not meet the standards of appropriate waste management systems. Poor waste management leads to serious environmental issues at the local and global levels, for which an effective and sustainable waste disposal system is in urgent need. Due to its proven waste degradation and biotransformation capabilities, the Black Soldier Fly (BSF) provides a potential and economical alternative to recycling biological waste. The current study investigated the interactions between municipal and organic waste with the help of Black Soldier Fly Larvae (BSFL) as a growing medium for substrate culture by comparing the physicochemical parameters of waste before and after BSFL treatment of municipal and organic waste. The study results revealed that BSFL can improve the quality of the final product and promote the degradation of organic waste, although BSFL cannot effectively directly degrade municipal solid waste, it can reduce municipal solid waste by efficiently degrading organic waste. The optimal environmental conditions for BSFL breeding and growth was summarized in this study, and the environmental conditions of four seasons of four representative cities in different location of Pakistan were collected and analyzed. The most suitable cities for the development of the BSFL waste treatment system were first inferred, which makes basic research for developing an economical and feasible waste treatment system in Pakistan. A comprehensive legal framework regarding integrated MSW management systems should be introduced and implemented in Pakistan, including that local municipal authorities should collect the waste collection fee.



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## 1. Introduction

The acceleration of economic growth in the 21st century results from industrial civilization, which has changed all countries in the world. The United Nations Department of Economic and Social Affairs reported that the world's urban population accounted for 55% of the total population in 2018. It is expected to rise to 68% by 2050, and more than 90% of the population growth will occur in Asia and Africa (Akmal and Jamil, 2021). The fast growth of population, urbanization, and economic development in countries that require high-quality foods and managing organic waste will need many resources (FAO, 2017; Surendra and Kuehnle, 2019). To fulfill these resources, agriculture production should be increased up to 49% between 2013 and 2050 (FAO, 2017). Due to unsustainable production and consumption patterns such as lifestyle changes, shortened product life spans, and linear economy, waste accumulation is receiving increasing attention on a global scale. Research estimates that poor waste management and improper disposal introduce 8 million metric tons (Mt) of plastic into the ocean every year (Lau et al., 2020). Organic waste is produced in huge amounts worldwide, especially in developing countries, and is not managed properly (Troschinetz and Mihelcic, 2009; Kaza et al., 2018; Wilson et al., 2012). As compared to former waste streams, there are many other causes, such as biological instability as a result of biodegradation and enzyme activity, high nutrients and moisture contents, and the potential existence of pathogens, which make organic waste management more stimulating (Varelas, 2019; Russ and Meyer-Pittroff, 2004; Yin et al., 2014). In general, economic, environmental, food security, and public health problems are related to the increasingly unsustainable ways in which limited resources are used to produce food. Besides generating organic waste in massive amounts, there is also a need for a more technically feasible, economically viable, and environmentally friendly approach to food and feed production and organic waste management.

It is estimated that Pakistan generates 87,000 metric tons per day (TPD) of municipal solid waste (MSW) every day, with an annual growth rate of 2.5% (Aslam et al., 2021). Insufficient waste collection and recycling can cause adverse environmental impacts and public health challenges (Aslam, 2020). As a result of poor municipal solid waste management, the major cities of Pakistan are in the worst condition. Organic waste is a major part of MSW (Muhammad and Zakariya, 2013). Socioeconomic development, population, and income levels often affect the production of MSW (Wilson, 2007; Alabaster et al., 2012). All cities in Pakistan produce a large amount of MSW, with an annual growth rate of 2.4% (Environmental Protection Agency, 2005; Adeel et al., 2012). The only option for the management of MSW is being considered open dumping. In contrast, the country faces a serious power shortage. Therefore, like in other developing countries, most non-renewable energy sources remain the choice in Pakistan (Pakistan Economic Survey, 2013–14). In this sense, converting waste into energy will be the most reasonable way and can be used as a more attractive method on a global scale. The shortage of skilled labor, insufficient policy and institutional support, and inadequate technical and financial resources are the main obstacles to proper solid waste management (SWM) in Pakistan (Akmal and Jamil, 2021).

Karachi, Pakistan's largest and most populous metropolis, is located in South Asia and is the country's largest city. The city produces 12000 tons of MSW every day, with 52% of it being organic (Mahmudul et al., 2021). In Pakistan, there is currently no effective waste management system. Large cities like Karachi, where vast MSW are produced and mistreated, resulting in serious environmental and public health problems, are particularly at risk (Zhang et al., 2021). MSW is collected and openly disposed of in so-called landfill sites outside Karachi. In contrast, 40% of the waste is left on the streets and roadsides, creating a hazardous environment (Pang et al., 2020). A significant amount of MSW is generated from industrial zones in Karachi, as in other major cities in Pakistan (Mahmudul et al., 2019). A lack of waste management and control means that this resource is wasted. Therefore, it is becoming a matter of health and environmental issues (Maldonado-Alameda et al., 2020).

Similar is the case of Pakistan's other major cities, i.e., Islamabad, Lahore, and Peshawar, which is a huge task. The second-largest city in Pakistan is Lahore (Irfan et al., 2020). As a result, many MSW facilities have exploded in recent decades (Azam et al., 2020). Due to its superior infrastructure, facilities, and collecting efficiency of MSW, the Lahore Waste Management Firm was chosen as a representative treating company in this study (Ferronato and Torretta, 2019). Because all of the major cities in the country experience the same seasons simultaneously, the factors influencing MSW generation (such as geographic location, industrial status, infrastructure, and socio-culture) are less likely to make any discernible distinctions across the cities (Saber et al., 2021). As a result, the categorization of MSW in Lahore can be utilized as an excellent reference to describe the waste management status (Sadef et al., 2016). Peshawar has an estimated population of four million and is considered one of the largest cities in Pakistan. Like other major cities in Pakistan, Peshawar also faces the problem of solid waste management, as the city generates between 600 and 700 tons of municipal waste every day, with a daily production of approximately 0.3–0.4 kg (Yaman, 2020). Waste generation rates per capita vary significantly in different regions of Peshawar because the majority of the population lives in low- and middle-income areas (Afridi and Qammar, 2020). Unacceptable air quality in the city is caused by a lack of municipal solid waste collection and disposal services, as around 60% of solid wastes linger at collection locations or in the streets. Hazar Khwani and Lundi Akhune Ahmed are currently being used for open dumping. Tautened of individuals in the city participate in waste scavenging (Azam et al., 2020). Scavengers burn solid rubbish in open dumps to recover recyclables, including plastics, glass, and metals, which is an alarming and dangerous practice. Residents' recycling rate fluctuates and cannot be retrieved by authorities without purchasing directly from the residents. This can only be done with recyclables that make it to a specific bin or secondary transfer station (Venâncio and Pope, 2018).

Poor management of municipal solid waste in Gilgit, Pakistan, has lost most of the city's attractiveness. Gilgit Municipal Committee (MC) collects and transports solid garbage. Trash collection is a big part of waste management, while the other half is left on the streets and in residential neighborhoods. Trash is removed from the streets and sidewalks by sweepers. Collection and disposal of solid wastes are entrusted to the GMC. Near Chilmish das, Gilgit has only one authorized MSW waste disposal site. That is because there is no suitable way to conceal this rubbish. (Saber et al., 2021). Waste is thrown in the open since there are no facilities to treat it properly. Poor waste management has resulted in the introduction of several viruses and vector-borne diseases, resulting in environmental pollution. No mechanism exists to recycle this garbage (Yasin and Usman, 2017).

To elucidate the problem of municipal solid waste management in Pakistan's major cities, a suitable solution is highly recommended. In addition to chemical treatment, anaerobic digestion, composting, vermicomposting, and other techniques can biodegrade organic waste. Unfortunately, vermicomposting cannot happen because of low bioconversion rates and other environmental factors that limit the worm's activities. Commercially, anaerobic digestion is not very popular since it requires control conditions that are difficult to develop on a large scale (Kumar et al., 2018). People have become more and more interested in applied entomology in recent decades. Insect predators have always attracted attention while being used for the biological control of pest species in agricultural ecosystems, but now the emphasis is being placed on insects to develop biologically inspired technologies as alternative food sources (da Silva and Hesselberg, 2020). Since the first introduction of waste treatment in the 1990s, the black soldier fly (BSF) has received increasing attention as an effective method of bio-waste conversion (Mertenat et al., 2019). The black soldier fly (BSF) (Diptera: Stratiomyidae) is considered the "crown jewel" of the fast-growing insect culture industry (Kaya et al., 2021). The larvae of BSF can grow in a variety of decomposable organic waste streams, like mixed crop debris, human feces, and animal manure. The degradation efficiency of these decomposable organic wastes by BSF larvae ranges from 55 to 80% (Beesigamukama et al., 2021). Black soldier fly (BSF) larvae are exciting because they provide potential and economical alternatives for recycling biological waste (da Silva and Hesselberg, 2020). Its habitat can be found widely in warm temperatures and tropical and subtropical areas (Guo et al., 2021). This species is multicultural distributed in warm temperate and tropical regions (approximately between 45°N and 40°S) ranging from the Neotropics to the Australasian, Nearctic, Palearctic and Afro-tropical Regions. However, the species was originally reported by a Hilo Sugar Company in the Hawaiian Islands and is native to the United States (Singh and Kumari, 2019).

BSF larvae encourage a high rate of waste biomass conversion. For example, it has been reported that the initial capacity of kitchen and municipal waste has been reduced by 50%–80%, and the biomass yield of BSF is higher (approximately 15% as compared to the larval weight). The Larvae of BSF are also quite effective in handling wastes contaminated by heavy metals or pesticides. BSF larvae greedily prey on organic waste, like rotting plants or animal organs, animal or human manure, food waste, slaughterhouse waste, distiller grains, sludge, etc. (Singh and Kumari, 2019). According to reports, the phosphorus and nitrogen contents of waste are significantly reduced to 85% and 75%, respectively. Due to releasing carbon dioxide ($CO_2$) into the atmosphere, this technology has some global warming potential. However, the former has been concealed due to its many advantages and ability to promote carbon sequestration. Compared to other traditional composting methods, the global warming potential of BSF technology is minimal (Salomone et al., 2017). Black soldier larvae have high proteins, lipids, polysaccharides, calcium, etc. (Popa and Green, 2012). In the larval stage, the black soldier fly can reduce fecal waste by 50% (Nguyen et al., 2015). BSF larvae, after waste treatment, are rich in nutrients, averaging 42.1–43.2% crude protein, 33% fat, and micronutrients. Therefore, they are promoted as a suitable substitute for fish or soybean meals in poultry, pig, and fish feed and provide income-generating opportunities (Chia et al., 2018a,b). Diptera can feed voraciously on various waste streams and convert organic composites into larval biomass, which can be used as animal feed. This capacity can be harnessed in the circular economy for recycling organic food waste and mitigating the adverse effects caused by its release into the environment (Dzepe et al., 2021).

The current study investigated the interactions between municipal and organic waste with the help of BSFL as a growing medium for substrate culture. Specifically, this study aimed to evaluate (1) which season in Pakistan is suitable for BSFL technology implementation. (2) The effects of environmental variables on the performance of BSFL for degradation of waste in Pakistan. (3) Would BSFL be a good nutrient source for substrate culture; and subsequently, determine (4) the optimal conditions of BSFL to be used in the biodegradation of waste. Municipal and organic waste samples were collected from the study area to determine the direction/geographical location of Pakistan and which session is feasible for biodegradation of waste by BSFL technology. Although the species thrives in warm tropical regions, only a few studies have been carried out in Pakistan, such as Salam et al. (2021) research on waste management of BSF larvae, without much attention to its role. In this study, we explored the role of BSF in waste management. The study also sought to provide knowledge and advice on how the BSFL treatment facility can best operate in different types of waste in the north, south, east, and west directions of Pakistan.

## 2. Methodology

### 2.1. Study area

The current research study was carried out in Pakistan, a country in South Asia, from 2020 -to 2021. Pakistan has a highly diverse geography and climate. The monsoon seasons with frequent flooding due to enormous rainfall, while it has a dry time of year with significantly little precipitation or none at all. Pakistan consists of four seasons: the winter season from December to February, the spring season from March to May, the summer season from June to September, and

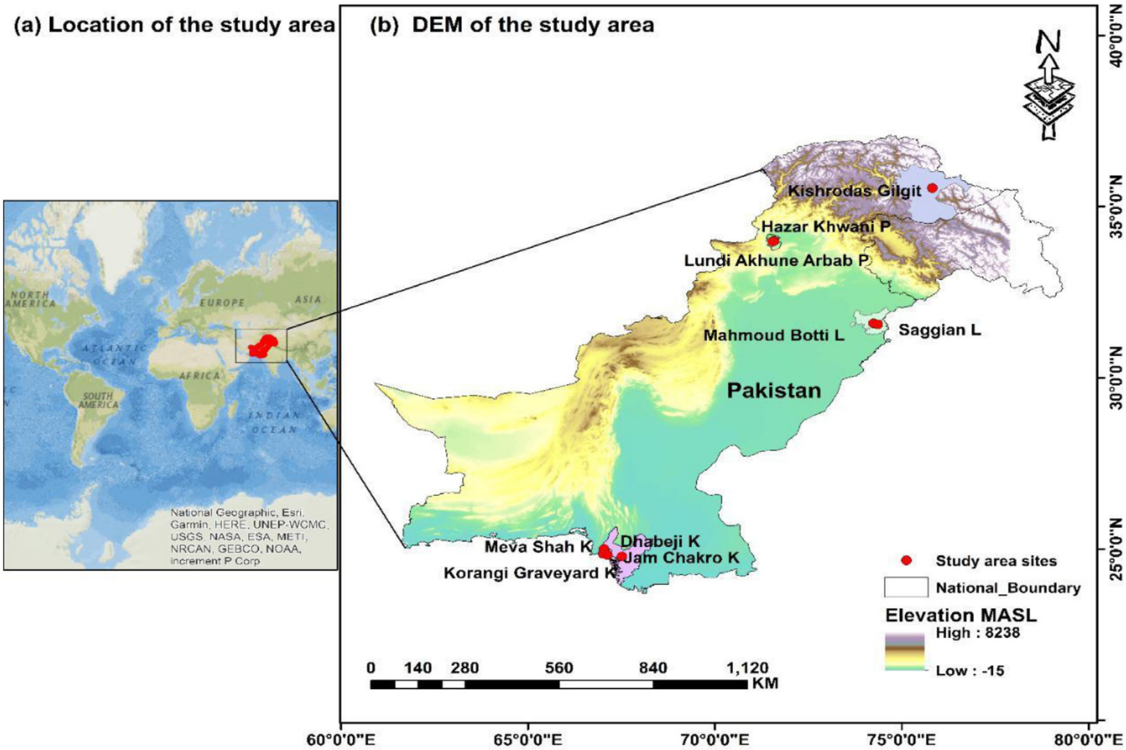


**Fig. 1.** GIS map showing the selected locations.

the monsoon season from October to November. Precipitation pattern significantly fluctuates every year, and alternative flooding patterns and drought are common. Presently an extreme variation in climate is experienced by the cities in (North, South, East, and West) with very high temperatures in summers; however, the relative humidity levels are significantly low in the range between 30 and 80% in Pakistan. The country experiences tropical to temperate conditions, with arid conditions on the southern coast with an average annual rainfall of 408.3 mm in 2020 and 25.64 mm in 2021.

The study was carried out in Gilgit, Lahore, Peshawar, and Karachi. We chose these big cities in Pakistan because the quantity of waste created in them is much greater than in other cities. These cities do not have proper waste management methods to treat the daily waste generated. Therefore, being a developing country, Pakistan does not have the economy to manage waste. However, BSF is an alternative and cost-effective method to treat solid waste. In the future, for the establishment of BSF industries, these cities are recommended. If BSF industries are established away from these cities, it will require cost to transport waste from the cities to the industry.

The administrative capital of Gilgit-Baltistan is at the confluence of the Gilgit and Hunza rivers. It is situated between 34-04° North latitude and 72-30° to 77-50° East longitude. The city extends longitudinally from north to south along the Gilgit River. The village is located 1490 meters above sea level. The administrative capital of Gilgit-Baltistan is located at the convergence of the Hunza and Gilgit rivers. It is located between 34-04° north latitude and 72-30° to 77-50° east longitude. The city stretches along the Gilgit River from north to south. The height from the sea level is about 1490 m. The average annual rainfall is 165.4 mm, with a minimum temperature of −8.3 °C and a high temperature of 23.6 °C. While Lahore has extremely hot summers and cool winters, it lies in a semi-arid climate. There is a monsoon season between July and September. Peshawar is the capital of Kp province. It is the largest city in Kp province. The average population of Peshawar is about 4.2 million people. The people of Peshawar enjoy four seasons because of their geographical location. During summer mean temperature is 42.7 °C, while in winter, the temperature reaches 2.5 °C. Karachi, one of the biggest cities in Pakistan, is located on the seaside of Sindh province. Karachi also experiences four seasons in the year, but this city does not have a high temperature in summer and is not so low in the winter season. Fig. 1 shows the map for the selected locations.

### 2.2. Black Solider Fly Larvae (Hermetia illucens)

#### 2.2.1. Species distribution based on geography

*Hermetia illucens* is one of five genera that belong to the subfamily Hermetiinae. The fly belongs to the order Diptera. The other four genera are *Chaetohermetia, Patagiomyia, Chaetosargus,* and *Notohermetia*; BSFL is the most widely scattered

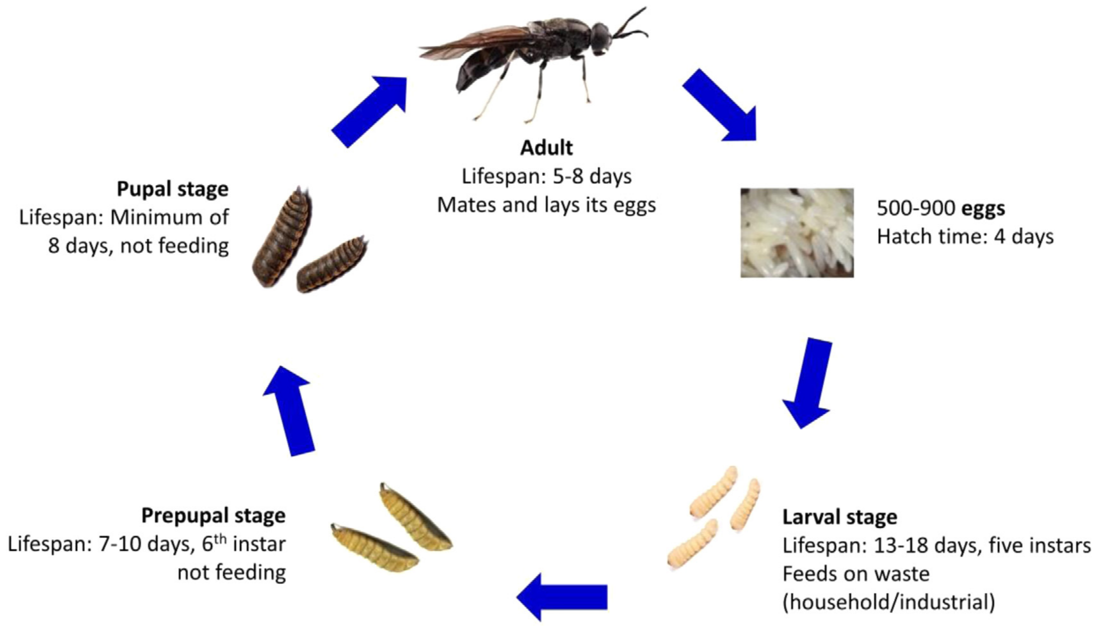


**Fig. 2.** Life cycle of Black Soldier Fly Larvae.

species in the world (Zhang et al., 2021). This species is widely distributed in warm temperate and tropical regions (between 40°S and 45°N), ranging from the neotropics to Australasia, the Palearctic, and areas of Afro-tropics. However, the species was originally reported by a Hilo Sugar Company in the Hawaiian Islands and is native to the United States (Song et al., 2021).

### *2.2.2. Life cycle with basic anatomy*

These flies are large, thin, black species with brown wings and tentacles in three segments projected from the head. The antennae are composed of 8 flagella; the former is the size of the antennae (Setti et al., 2019). Like other Diptera species, the body of an adult fly is separated into the head, chest, and abdomen. The eyes are bare and wide apart. The abdominal segment consists of 5 parts that have white abdominal spots (Singh and Kumari, 2019). Female flies are typically shorter than male flies. The length of the body and wings of the female usually ranges from 12 to 20 mm and from 8 to 14.8 mm (Surendra et al., 2020).

The life cycle of BSFL consists of 5 stages, i.e., egg, larvae, pupae, pre-pupae, and adult (Fig. 2). The entire life cycle is mainly contributed by the larval and pupal stages, while the relatively short stages of BSFL include the adult and egg hatching stages (Kaya et al., 2021). As the BSFs life span is very short, the female larvae will lay a considerable amount of eggs, which will develop into newly hatched larvae very soon (Liu et al., 2021). Depending on the environmental conditions, the life cycle of these flies is very small, approximately 6–8 days, although it can be extended to 6–7 weeks, as shown in (Table 1) because these flies can slow down their activities under contrary situations (Parodi et al., 2021). In inadequate food supply and low-temperature conditions, these species can achieve a flexible life cycle by prolonging the larval stage (approximately 4 months). This phenomenon helps the larval population's feasibility during the organic waste shortage periods (a food source for newborns). However, egg production and larval development are highly dependent on food quality (Song et al., 2021).

There will be only one oviposition event in the entire life cycle of adult females because they mate only once. The oviposition event occurs in dry crevices near the food source because mating occurs two days after emergence (Rehman et al., 2019). After two days of mating, the adult female lays eggs and quickly hatch into newly hatched larvae (Sarpong et al., 2019). Since most of the nourishing occur at this stage, the larvae will soon become the last immature stage, that is, the prepupae and feeding stops. Prepupae migrate from food sources to dry pupal crevices (Guo et al., 2021; Wang and Shelomi, 2017). At this stage, the species reaches its maximum size, containing 36%–48% protein and 33% fat, and finally, when the adult worm develops at 14 days, it completes metamorphosis. Adult flies do not have a mouth, digestive system, or poisonous stingers, so they do not threaten living organisms (da Silva and Hesselberg, 2020; Dzepe et al., 2021).

## *2.3. Factors affecting the growth and development of black soldier fly larvae*

The growth and development of BSFL are affected by several environmental parameters like temperature, humidity, sunlight, moisture contents, pH, etc.

**Table 1**
Life history of Black soldier fly (under optimum condition) for biodegradation of waste.

| Sr. no | Black Solider Fly Phenomena | | |
|---|---|---|---|
| 1 | Duration | 12–15 days | Grimaldi et al. (2020) |
| 2 | Conversion agent | Black soldier fly larvae | Mazza et al. (2020) |
| 3 | Optimal temperature | 30 °C | Chia et al. (2018a,b) |
| 4 | Optimal moisture content | 65%–90% | Liu et al. (2021) |
| 5 | PH | 6.0–8.0 | Ma et al. (2018a,b) |
| 6 | Humidity | 60%–65% | da Silva and Hesselberg (2020) |
| 7 | Sun Light intensity | 80% mating occur at 110 μmol $m2^{s-1}$ | Singh and Kumari (2019) |
| 8 | Applicable feedstock | All most everything | Beesigamukama et al. (2021) |
| 9 | End product | Humus, biodiesel, protein, sugar | Mohsenizadeh et al. (2020) |
| 10 | State of the end product | Require maturation treatment | Gold et al. (2020) |
| 11 | Gas emission | Completely aerobic process and nil emission | Gholib et al. (2020) |
| 12 | Additional benefits | Suppress the growth of Domestic Musca | Fadhillah and Bagastyo (2020) |
| 13 | Cost | 6–16 USD/ton. | Bortolini et al. (2020) |

#### *2.3.1. Humidity and temperature*

BSF is a eurythermic species that can tolerate a wide temperature range (15–47 °C), but at the same time, it is tremendously sensitive (Park, 2016). The different stages of BSFL are affected by increasing or decreasing temperature. Hatching eggs from BSFL larvae incubated at high temperature takes less time than at low temperature. The development time of the larvae differed significantly between temperatures and between rearing substrates. There is a significant interaction between the effect of temperature and the brood substrate on the development time of the larvae. The longevity of adult BSF is significantly affected by temperature in both females and males. Adult flies live longer at intermediate temperatures than at higher extreme temperatures (Chia et al., 2018a,b).

Relative humidity and temperature significantly affect the development, mating, and oviposition of species (Tomberlin et al., 2009; Barry, 2004). Several authors carried out related studies and concluded that most (approximately up to 99.6%) oviposition occurs in a temperature range of 27.5 to 37.5 °C and relative humidity of 60% (Booth and Sheppard, 1984; Holmes, 2010; Sheppard et al., 2002). Other studies by many authors have shown that maintaining a relative humidity of around 60% and a temperature of 27 °C at the site are perfect conditions for egg-laying and mating (Sheppard et al., 2002; Holmes, 2010). The research results are also positively correlated with each other.

#### *2.3.2. Sunlight*

Direct sunlight plays a significant role in mating, survival, growth, and development in the natural environment. Artificial light sources are essential for indoor experiments. Sunlight with an intensity of 110 μmol $m^2$ $s^{-1}$ is required for almost 85% of mating events (Park, 2016). In one study, artificial light sources were reported to impact species' breeding events. This precise method is beneficial and effective for raising species outside their natural habitat, where the primary source of influence is the sunlight. The mating and oviposition of the species are positively correlated with quartz iodide lamps (135 μmol $m^2$ $s^{-1}$) as sources of artificial light (Zhang et al., 2010). Insects cannot see the light beyond 700 nm, so the wavelength range between 450 and 700 nm is best suited for adult breeding events (Briscoe and Chittka, 2001; Zhang et al., 2010).

#### *2.3.3. Moisture content*

Moisture contents of the waste significantly influence the survival and growth of BSFL (Cheng et al., 2017). Excessive moisture content obstructs the decomposition rate and is agglomerated, and thick materials can accompany the subsequent residue, making further processing difficult (Diener et al., 2011b). Proper control of moisture contents can help overcome these issues. Fatchurochim et al. (1989) concluded in their research study that when the moisture content is between 40% and 60%, the survival rate of flies fed with poultry manure is highest. Similarly, Banks (2014) revealed in his study that the growth rate of pupae is higher when 85% of the water in the fecal sludge is present.

#### *2.3.4. pH*

The value of pH is an essential parameter that influences the survival and life span of BSF. Various research studies on black soldier flies have revealed that the optimum condition for larval development and growth is a pH higher than 6 (Meneguz et al., 2018). Green and Popa (2012) revealed that the larvae of BSF can adjust the liquid organic leachate pH up to 9. Alattar (2012) also determined that the ability of the organic medium to adjust the pH strictly depends on the density of the larvae. Ma et al. (2018a,b) further studied the influence of different pH levels on the development of larvae and illustrated that the larval growth performance is high at pH values above 6 to 10. The larval body weight is increased at pH levels of 4 or 2. It was also determined that the larvae in alkaline substrates could adjust their pH value from 8 to 8.5 but could not adjust the pH value of strongly acidic substrates.

### 2.4. Optimal conditions for the survival of Black Solider Fly

Like other flies, BSF is sensitive to several environmental parameters, predominantly temperature, which is the most important non-biological factor. Not only can it affect the development rate, season, and insects' daily cycle, but it has a significant effect on the different features of insect biology, such as adult life expectancy, larvae survival, fertility, growth, sex ratio, and population growth parameters (Chia et al., 2018a,b). The two fundamental factors which affect the black soldier fly's foraging, development, and life cycle are temperature and humidity. The BSF has been used for waste management purposes in low latitude areas, where the temperature and sunlight are very suitable for the species to reproduce throughout the year (Alvarez, 2012). Moisture content plays an essential role in composting because it is a significant parameter for the survival of microorganisms, especially for the growth and development of BSF. The optimal water content of the feed varies between 65% and 90%, as shown in Table 1 (Liu et al., 2021).

The pH value is an inherent parameter that affects the life cycle and survival of BSF. Many studies on the black soldier fly revealed that a pH higher than 6 is the best condition for the survival, growth, and development of larvae. As shown in Table 1, it is recommended that BSF be used for organic waste biotransformation with an initial pH of 6.0 to 8.0 (Ma et al., 2018a,b). The initial pH effect may be favorable to beneficial bacteria, thereby contributing to the survival, growth, and development time of the larvae; the gut microbiome of insects promotes weight gain, growth, and egg production of larvae (Ma et al., 2018a,b). Sunlight plays a key role in the mating events of the species in the natural environment. The study results demonstrated that sunlight with an intensity of 110 $\mu$mol $m^2$ $s^{-1}$ is required for almost 85% of mating events, which is the leading cause why mating in the winter season is highly restricted (Singh and Kumari, 2019).

### 2.5. A climatic sampling of data

The data for this study was collected in 4 different seasons of Pakistan, i.e., autumn, winter, spring, and summer, from the East, West, North, and South directions of Pakistan. Different weather parameters were measured, such as temperature, humidity, sunlight, and rainfall. In all seasons, weather parameters were collected from Gilgit, Karachi, Peshawar, and Lahore on the North, South, East, and West sides of Pakistan. The autumn season in Pakistan is started in September and ends in October, and the weather data was collected in autumn 2020.

The weather parameters for these cities were measured for the winter season from winter 2020 -to winter 2021. The winter season starts in November and ends in February. Then, the spring season lasts for two months, March and April. The weather data was collected from May 2021 to August 2021 from these cities in the summer season. The temperature, humidity, sunlight, and rainfall were measured three times during all seasons, at the start of the season, middle of the season, and end of the season. The minimum, maximum, and average temperatures were measured using a Thermometer. The humidity and rainfall of the selected cities were measured by a Hygrometer and rain gauge.

### 2.6. Analytical method

This study characterized two types of waste: municipal waste and organic waste. These wastes were analyzed by measuring their different parameters in the North, South, East, and West direction of Pakistan. The waste parameters were pH, temperature, electrical conductivity, moisture content, and C: N. The waste parameters of the selected four cities, Gilgit, Karachi, Lahore, and Peshawar, were analyzed using different devices for their measurement. The pH and temperature of organic and municipal waste were measured using an electric pH meter and Smart sensor AS 89000. The electrical conductivity and moisture content were measured by an HI 8633 electrical conductivity meter and moisture meter.

The method for determining DOC content is to extract fresh samples with ultrapure water at room temperature at a ratio of 1:10 (w/v) with a 180-rpm shaker for 24 h, and then filter the total organic carbon (TOC) Analyzer (TOC-LCPH, Shimadzu, Japan) with a 0.45 $\mu$m nylon syringe filter. Determine the parameters of total Kjeldahl nitrogen (TKN), total phosphorus (TP), total potassium (TK), and organic matter (OM) by air-dried samples. To determine TKN, TP, and TK, 0.5 g of dried and sieved samples were digested with 10 mL $H_2SO_4$ and 1 mL $HClO_4$, TKN was determined by the Kjeldahl nitrogen analyzer (KDY-9820), and TP was determined by molybdenum–antimony colorimetry (TMECC, 2002). The value of OM was determined by cremating the sample in a muffle furnace at 550 °C for 24 h (Liu et al., 2021).

### 2.7. Statistical analysis

All statistical data were repeated 3 times and confirmed by the analysis of variance test (ANOVA) ($p < 0.05$), and the figures were drawn by origin 2016 (Origin Lab, America). The correlation analysis between statistics was determined using SPSS software (22nd edition).

## 3. Results

### 3.1. Performance of BSFL degradation on municipal and organic waste

Based on the current investigation, it was evaluated that the BSFL's ability to bio-transform the physicochemical properties of the two types of wastes into organic waste and municipal waste through the final treatment of residues.

**Table 2**
Characteristics of waste material used in this study.

| Parameters | Municipal waste | Organic waste |
|---|---|---|
| Moisture content (%) | 85.12 ± 0.08 | 81.35 ± 0.22 |
| pH | 6.36 ± 0.28 | 7.02 ± 0.07 |
| EC (μS/cm) | 153.7 ± 19.7 | 212.0 ± 47.78 |
| DOC (g/kg) | 0.445 ± 0.35 | 19.98 ± 4.60 |
| OM (%) | 59.69 ± 0.20 | 98.08 ± 0.13 |
| TKN (g/kg) | 21.27 ± 0.46 | 27.82 ± 0.24 |
| TP (g/kg) | 8.30 ± 1.36 | 2.30 ± 0.02 |
| TK (g/kg) | 2.69 ± 1.17 | 1.15 ± 0.39 |
| C/N ratio | 17.43 ± 0.78 | 21.46 ± 0.34 |

DOC — Dissolved organic carbon, EC — Electrical conductivity, OM – Organic matter, DOC — Dissolved organic carbon, TP — Total Phosphorous, DOC — Dissolved organic carbon, TK — Total potassium. Results are the average of three repeats ±standard deviation.

**Table 3**
Climatic conditions of the selected study area in the summer season of Pakistan.

| Summer season | Lahore | | | Gilgit | | | Peshawar | | | Karachi | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Temp (°C) | Humidity (%) | Rainfall (mm) | Temp (°C) | Humidity (%) | Rainfall (mm) | Temp (°C) | Humidity (%) | Rainfall (mm) | Temp (°C) | Humidity (%) | Rainfall (mm) |
| May | 40 | 35 | 16 | 3 | 60 | 26 | 36 | 43 | 48 | 35 | 63 | 6 |
| June | 39 | 47 | 42 | 10 | 50 | 10 | 38 | 43 | 56 | 34 | 64 | 7 |
| July | 35 | 67 | 148 | 12 | 49 | 9 | 36 | 61 | 218 | 33 | 67 | 40 |
| August | 40 | 53 | 119.6 | 10 | 50 | 64.2 | 35 | 51 | 250.4 | 30 | 75 | 161.5 |

In addition, it was tried to use BSFL to find changes in the correlation between the analysis parameters in organic waste and the degradation of municipal waste. BSFL can realize the utilization and recycling of organic waste and convert it into bio-fertilizer, but according to (Table 2), it is ineffective for municipal waste recycling. The characteristics of municipal waste and organic waste are shown in (Table 2). Furthermore, the study results indicated a significant difference between municipal and organic waste. The highest TK, TKN, TP, and GI were observed among the two treatments in the organic waste treatment. At the same time, the reduction rate of OM and moisture in organic waste treatment was also the highest. Compared with the organic waste group, the trend of all parameters in urban waste treatment is stable. Although the initial parameters are integrated, the concentration of DOC in municipal waste is 0.445 g/kg, which cannot provide a good carbon source for BSFL. However, the survival rate of BSFL degraded organic waste was 85%, while the survival rate of BSFL degraded municipal waste was 46%. These results indicated that BSFL could improve the quality of the final product and promote the degradation of organic waste. BSFL biotransformation can perform valuable functions in the form of organic waste management and produce useful products in the form of bio-fertilizers.

Rehman et al. (2017) also reported similar results when adding BSFL and soybeans to dairy manure compost. The results of the OM decrease trend are consistent with the previous studies. The previous report showed that the OM decreased after BSFL composted different types of organic waste. Lalander et al. (2018) reported a similar result that BSFL is not sensitive to sewage sludge. The decrease in OM may be attributed to the early conversion of OM into VFA and its use as feed by BSFL (Mertenat et al., 2019). The carbon lost in the form of $CO_2$ through respiration is the cost of OM reduction. OM degradation can be considered a parameter of raw materials and their conversion into stable products (Sharma and Garg, 2018).

### *3.2. BSFL survival rate with climatic conditions in summer seasons of Pakistan*

Table 3 presents the weather conditions of four different cities, Lahore, Gilgit, Peshawar, and Karachi, in the summer season. The highest temperature was recorded in May 2021 in Lahore, which was 40 °C, and the lowest temperature was 3 °C recorded in Gilgit in May 2021. The other three cities, Peshawar and Karachi, showed temperatures of more than 30 suitable for BSF to survive. The highest and lowest humidity was 70 and 35% in Karachi and Lahore. As BSF needed 60%–65% humidity (da Silva and Hesselberg, 2020). Therefore, the Humidity level of Karachi is best for BSF. In summer 2021, among these four cities, the highest rainfall was in the order Peshawar > Lahore > Karachi > Gilgit. The moisture content required for BSF is 65%–90%. According to the rainfall situation in different cities of Pakistan, BSF can survive in Peshawar, Lahore, and Karachi as the rainfall can moisten the waste and provide a better place for BSF to develop (Liu et al., 2021).

BSF is a eurythermal species that can tolerate a wide temperature range (15–47 °C), but at the same time, it is susceptible. The black soldier fly is a tropical and warm-season temperate species (Sheppard et al., 1994). Adults have a high survival rate at 27 °C and 30 °C (Caligiani et al., 2019). Temperature and relative humidity significantly affect the development, mating, and spawning of species. Studies conducted by multiple authors have shown that 99.6% of

**Table 4**
Climatic conditions of the selected study area in the Autumn season of Pakistan.

| Autumn season | Lahore | | | Gilgit | | | Peshawar | | | Karachi | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Temp (°C) | Humidity (%) | Rainfall (mm) | Temp (°C) | Humidity (%) | Rainfall (mm) | Temp (°C) | Humidity (%) | Rainfall (mm) | Temp (°C) | Humidity (%) | Rainfall (mm) |
| Sept | 37 | 38 | 85.1 | 6 | 48 | 69.8 | 34 | 38 | 18.7 | 30 | 69 | 1.9 |
| Oct | 34 | 16 | 0 | −2 | 54 | 38.1 | 31 | 21 | 0 | 32 | 46 | 1 |

**Table 5**
Climatic conditions of the selected study area in the Winter season of Pakistan.

| Winter eason | Lahore | | | Gilgit | | | Peshawar | | | Karachi | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Temp (°C) | Humidity (%) | Rainfall (mm) | Temp (°C) | Humidity (%) | Rainfall (mm) | Temp (°C) | Humidity (%) | Rainfall (mm) | Temp (°C) | Humidity (%) | Rainfall (mm) |
| Nov | 24 | 24 | 5.4 | −14 | 65 | 6.2 | 21 | 28 | 43.9 | 28 | 38 | 0 |
| Dec | 19 | 35 | 6.3 | −21 | 75 | 2.5 | 16 | 34 | 9.4 | 24 | 33 | 0 |
| Jan | 17 | 70 | 25 | −24 | 68 | 28 | 18 | 64 | 80 | 26 | 44 | 6 |
| Feb | 21 | 64 | 29 | −17 | 76 | 55 | 18 | 67 | 92 | 27 | 46 | 3 |

oviposition occurs in a temperature range of 27.5–37.5 °C and relative humidity of 60% (Gold et al., 2018; Singh and Kumari, 2019). Black Solider flies reproduce in warm temperatures, and almost all oviposition occurs at >26 °C (Cickova et al., 2015). Majeed et al. (2018) reported that when the temperature rose beyond 47 °C, the larval survival rate would fall rapidly. The sunlight intensity in summer is higher than in other seasons of Pakistan, and BSF also needs high. Therefore, summer is good for the production and development of BSFs.

### *3.3. BSFL survival rate with climatic conditions in autumn seasons of Pakistan*

In the autumn season of 2020, the highest temperature was recorded at 38 °C and 37 °C in Peshawar and Lahore, compared to the other two cities (Table 4). The highest to lowest humidity % was in order Karachi > Gilgit > Peshawar > Lahore. The quantity of rainfall was low in Karachi and Peshawar and high in Gilgit and Lahore. However, according to the literature, BSF biotransformation studies on human feces show ideal conditions are from 25 to 30 °C. Perfect breeding conditions have been observed at 32 °C, but temperatures of 15 to 47 °C can be tolerated.

However, the temperature and humidity in most studies are around 27 °C, and 67%, respectively, or the temperature is in the range of 20–35 °C, and the humidity is above 30%. The final weight of larvae raised at a temperature of 27 to 32 °C is 30% heavier than that of larvae raised at a higher or lower temperature, although they will take a little longer time for development at these temperatures (27–32 °C) (Bortolini et al., 2020). At 27 to 30 °C, females are 17%–19% heavier than males. Humidity affects the exclusion of BSF eggs. An environment with relative humidity (RH) of 70% may increase the survival rate of adult flies by 2 to 3 days, while low RH may increase the mortality of larvae (Silva et al., 2020).

### *3.4. BSFL survival rate with climatic conditions in winter seasons of Pakistan*

The winter season of Pakistan have minus temperature in all Northern areas; Gilgit is one of them. Hence, a black soldier fly cannot survive in the winter season in the Northern areas of Pakistan. It is reported earlier that the use of black Solider Fly may cause problems in temperate and cold regions. Adults are territorial and have complex mating behaviors, including courtship (Tomberlin and Sheppard, 2002). In addition, they need sunlight to mate (Čičková et al., 2014) successfully. In winter, the intensity of sunlight is very low throughout Pakistan. Therefore, the mating of BSF is difficult. Fig. 3 presents the weather conditions of four different cities in Pakistan in the winter season. The temperature of Lahore and Peshawar was low, but their rainfall amount and humidity were more moderate than Gilgit and Karachi, as shown in Table 5. The lowest recorded temperature was −24 °C for Gilgit, and the highest temperature was 28 °C for Karachi. In Karachi, there was no rainfall in November and December 2020. Peshawar city had more rainfall than Gilgit and Lahore. The average Humidity (%) was the same for Lahore and Peshawar and the highest for Gilgit. It was observed that there was very little difference in the humidity (%) of Lahore, Peshawar, and Karachi.

According to the literature, the temperature and daylight requirements are not available in colder climates or are significantly reduced (Alvarez, 2012). BSFs thrive in summer, and winter temperatures kill them, so they must leave, seek shelter in winter, or resettle each spring (Alvarez, 2012). However, considering that all eggs cannot hatch at these temperatures, lower (<15 °C) and higher (>40 °C) temperature development thresholds are fatal to insects. The viability and hatchability of the eggs observed in this study occurred between 15 and 40 °C. Our observations at the lower temperature threshold are consistent with the observations at 16 °C. The egg vigor levels at 15 °C and 40 °C are extremely low (<12%), which is consistent with the report at 16 °C. Therefore, the successful egg exclusion reported at 19 °C is

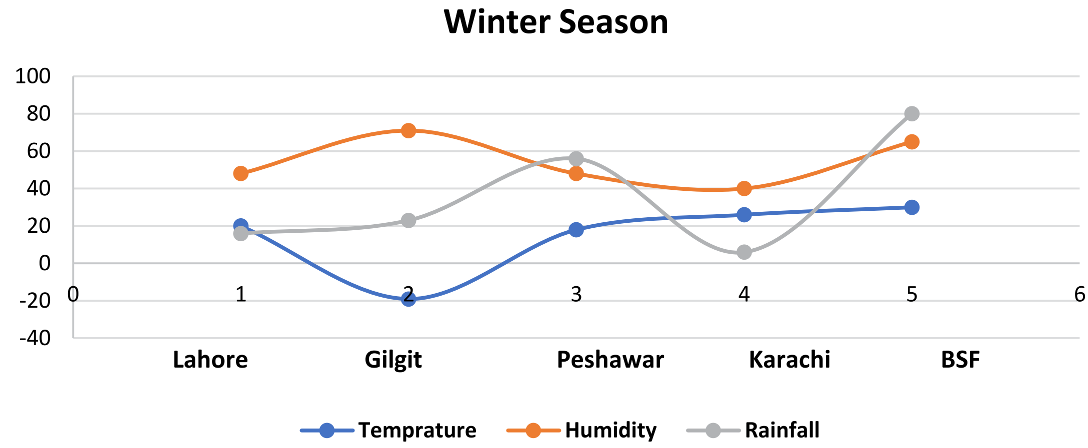


**Fig. 3.** Comparison between BSFL and Climatic conditions of the Winter season in Pakistan.

**Table 6**
Climatic conditions of the selected study area in the Spring season of Pakistan.

| Spring season | Lahore | | | Gilgit | | | Peshawar | | | Karachi | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Temp (°C) | Humidity (%) | Rainfall | Temp (°C) | Humidity (%) | Rainfall (mm) | Temp (°C) | Humidity (%) | Rainfall (mm) | Temp (°C) | Humidity (%) | Rainfall (mm) |
| March | 29 | 54 | 32 | −11 | 75 | 44 | 25 | 60 | 103 mm | 32 | 48 | 1 |
| April | 35 | 43 | 22 | −7 | 69 | 40 | 30 | 54 | 83 mm | 35 | 55 | 2 |

equivalent to the high egg exclusion rates recorded in this study at 30 °C and 35 °C, with survival rates of 80% and 75%, respectively (Chia et al., 2018a,b).

Although at a temperature of 27 °C, males and females are the most effective at all stages of development, at higher temperatures (30–36 °C), the metabolism increases, and the adults are still small, so they have a shorter life span and develop into a large extent restrict or inhibit. It was found that flies raised at 27 °C were about 5% heavier and about 10% longer than those raised at 30 °C. Further studies by many authors have shown that maintaining a relative humidity (approximately 60%) and temperature of 27 °C at the site are the best conditions for mating and egg-laying (Singh and Kumari, 2019).

Different temperature schemes produce suitability trade-offs within the acceptable development temperature range. For black soldier Fly, adults raised at 27 °C are about 5% heavier than adults raised at 30 °C and live about 10% longer. However, it takes an average of 4 days more to complete larval development at 27 °C than at 30 °C. Because adults do not eat except for water (Tomberlin and Sheppard, 2002), larval food is vital to health. Females spend more time in the larval stage than males and weigh more than adults. The trade-off is that although longer larval development may accumulate more energy reserves, thereby increasing fecundity (Tomberlin and Sheppard, 2002). The chance of obtaining a mate (Alcock, 1990), the risk of larval disease and parasitism, predation, resource degradation, and subsequent loss of mating opportunities may increase.

### *3.5. BSFL survival rate with climatic conditions in spring seasons of Pakistan*

In Pakistan, the duration of the Spring season is about two months, like Autumn, and it comes after the winter season. During this season, northern areas of Pakistan still have frozen temperatures, but the temperature has increased in other parts of the country. According to Table 6, the humidity level in Gilgit was more than in Karachi, Lahore, and Peshawar. The highest to lowest temperature in the four cities was in the order Karachi > Lahore > Peshawar > Gilgit. The highest temperature was 35 °C in Karachi and −11 °C for Gilgit. It was noticed that in every season, Gilgit has the lowest temperature, and in Spring and Winter, Karachi has the highest temperature. In Autumn 2020, Peshawar had 103 mm rainfall which was higher than the rate of rainfall for other cities, and Karachi had less rainfall among the four cities.

Black soldier flies generally adjust to temperate, tropical, and subtropical climates, however, only during the summer and spring months (April–October) (Holmes, 2010). Since the moisture content of the BSF diet plays a vital role in larval growth and survival, understanding the effects of the moisture content of food waste on bioconversion of BSF is vital for effective waste treatment of food (Cheng et al., 2017). Holmes et al. (2012) also previously concluded that 25% relative humidity would lead to a higher drying rate and high species mortality, but species with a relative humidity of 70% have a longer lifespan of 2 to 3 days. The growth and survival of the species are greatly influenced by the waste moisture

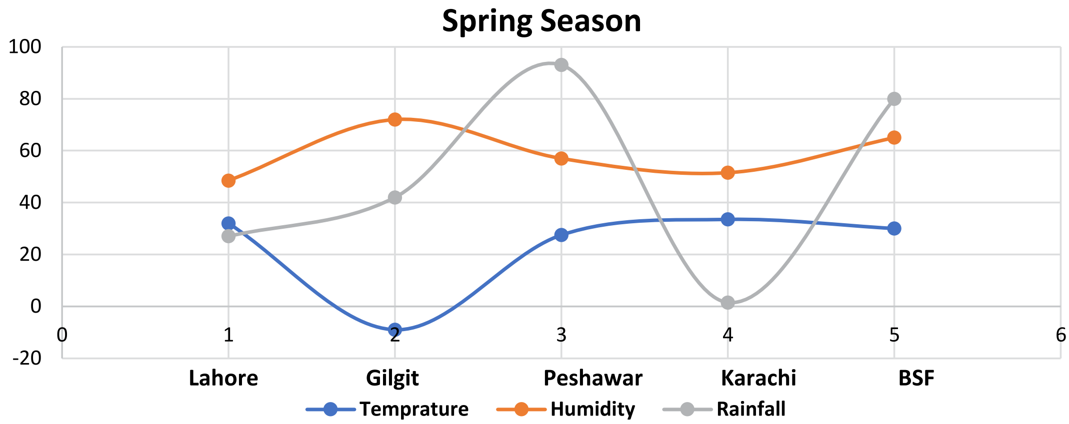


**Fig. 4.** Comparison between BSFL and Climatic conditions of the Spring season in Pakistan.

content (Cheng et al., 2017). According to (Fig. 4), Peshawar has more rainfall in Autumn 2020 than the other three cities. More rainfall will increase the moisture content of waste heaps. Hence, more rainfall is needed for the development of BSF. Peshawar provided the optimal temperature to BSF for their matting and development as the average temperature of Peshawar in Autumn is 27.5 °C and the humidity level of Peshawar was also more than Lahore and Karachi.

## 4. Discussion

Food waste treatment with BSFL has gained popularity due to its sustainability and potential use of BSFL-derived resources. Food waste treatment with BSFL can help reduce several environmental issues and toxic compounds generated by traditional disposal techniques like landfilling and incineration. BSFL may also be utilized to make essential items like oils and proteins. High-quality biodiesel may be made from bioconversion of food waste in many phases. BSFLs can also be used as animal feed because they have the same nutritional properties as normal food. The treatment of organic waste by *H. illucens* larvae has attracted more and more attention (De Smet et al., 2018; Henry et al., 2015; Gold et al., 2018; Cickova et al., 2015). In addition, other types of waste are being explored as potential substrates for sustainable use of *H. illucens* larvae biomass as food for humans and other animals (Chia et al., 2018a,b; Smetana et al., 2019). Previous studies have shown that the ecological characteristics of *H. illucens* are similar to those of the housefly, and the two species seem to compete with each other. *H. illucens* fly has been shown to inhibit the reproduction of house flies (Bradley and Sheppard, 1984; Sheppard, 1983). In 1997, Sheppard successfully developed the first fecal treatment system using *H. illucens*. In 1997, Dr. Oliver, an American environmental technology and engineering company, invented a bio transformer for black soldier fly larvae for household use. The bio transformer has been successfully used to treat municipal solid waste. In 2007, Dr. Xincheng from the Guangdong Institute of Entomology designed a bioconversion system for the efficient large-scale treatment of organic waste; this system achieved satisfactory waste reduction, decontamination, and recycling. Kitchen waste, livestock manure, spoiled fruits and vegetables, crop waste, and food processing waste can be quickly and efficiently transformed by *H. illucens* because of the wide range of potential food for this insect. The transformed waste then forms the biomass of *H. illucens*, so that its larvae can be used in environmental ecology and effective treatment of organic waste (Lohri et al., 2016; Hu et al., 2017).

Due to the rapid growth of population and economic activities and insufficient training in modern MSW management are responsible factors, waste management has become an emerging issue in Pakistan (Rehmani et al., 2020). The development of the MSW management system is also complicated (Ahsan et al., 2014). The average annual waste production in Pakistan is 2.4%, and the average per capita production per day ranges from 0.283 kg to 0.612 kg. Correct handling of MSW is a major issue in Pakistan (Korai et al., 2017). Existing BSF bio-waste processing facilities indicate that they are financially sustainable when processing several tonnes to several hundred tonnes of bio-waste per day with a high process performance (AgriProtein, 2018; Diener et al., 2015; Protix, 2018). Since BSFL treatment reduces the amount of biological waste, the concentration of heavy metals in the residue may limit its application as a soil conditioner or compost. Several studies have shown that inoculating biological waste with microorganisms can improve BSFL process performance (Xiao et al., 2018; Yu et al., 2011; Zheng et al., 2012).

BSFL can survive in harsh environmental circumstances, such as drought, oxygen deprivation, and food scarcity. For BSF adults, the optimal temperature ranges for mating and oviposition are 27–37 °C and 60%–70% relative humidity. The effectiveness of the composting process is improved by providing air, and we discovered that water content of 80%–90% is sufficient for BSFL feeding substrate. Changes in environmental circumstances significantly impact BSF larvae's growth and development. Temperatures ranging from 27 to 35 °C and humidity levels ranging from 70 to 75% are ideal for BSF larvae growth. Due to the photophobic nature of BSF larvae, dark ambient conditions are also necessary. The trade-off between larval growth rate and residue sieve efficiency was explored. Although a greater moisture content of the food waste leads to a faster larvae development rate, separating the residual from the BSFL biomass is problematic.

The recommended optimum range of 70%–75% for efficient residue sieving because sieving is no longer feasible at a moisture content of more than 80%. The studies on substrate pH have determined the impact of pH on larval development. Acidic (pH 2, pH 4, and pH 6) and basic (pH 8 and 10) conditions contribute to high BSFL survival. However, a pH range of 6 to 10 is more conducive for optimal larval growth and higher larval weight relative to a pH range of 2 to 4 because the alkalization of the substrates by larval activity stimulates protease activity, which increases the amount of protein available for larval growth. The BSFL bioconversion process raises the pH of the substrate and leaves an alkaline residue, making the substrate suitable for use as fertilizer. The ideal pH range of 7 to 8 stimulates plant growth and provides a favorable environment for nurturing beneficial bacterial populations in the residue.

Table 3 shows the summer weather in Lahore, Gilgit, Peshawar, and Karachi. Karachi's humidity is ideal for BSF. The BSF can live in Peshawar, Lahore, and Karachi because of the rain that moistens the garbage and provides a better environment for the BSF to flourish. According to Newby et al. (1997), the survival rate of larvae drops dramatically above 47 °C. Summer in Pakistan has more sunshine intensity than other seasons, and the BSF requires it. So, the summer is favorable for BSF production. According to the summer conditions in Lahore, Karachi, Peshawar, and Gilgit. BSF can best survive in Karachi due to its temperature and humidity as the Gilgit temperature is very low to survive, and humidity (%) of Lahore and Peshawar is also less than the required humidity, and the temperature of Lahore is tolerable but not the best to stay alive.

BSF requirement of temperature and humidity, Karachi is suitable for developing BSF among four cities (Table 4). Gilgit is completely unsuitable for the production and development of Black Solider Fly as the temperature of Gilgit is very low. But BSF can survive in Lahore and Peshawar. Compared to Lahore and Peshawar, the humidity level of Karachi is high in the Autumn season. The rainfall in Karachi is deficient than in the other three cities. Rainfall in the autumn season is low in Peshawar and Lahore than in other seasons of the year. In the Autumn season, the sunlight intensity is more in Lahore and Peshawar than in Karachi and Gilgit.

Winter in Pakistan brings cold temperatures to the north, especially in Gilgit. Consequently, a black soldier fly cannot survive the Pakistani winter. It was previously stated that using black soldier flies in cold locations may be problematic (Tomberlin and Sheppard, 2002). They require sunlight to mate (Cickova et al., 2015). During the winter months, Pakistan's sunlight intensity is low. BSF mating is difficult. The temperature in Karachi during the winter is suitable for the growth and mating of BSFs, but the humidity level in Karachi is too low, according to Table 5. In the spring season, northern areas of Pakistan still have frozen temperatures, but the temperatures have risen in other parts of the country. In Gilgit, the humidity level was higher than in Karachi, Lahore, and Peshawar. It was noticed that in every season, Gilgit has the lowest temperature, and Karachi has the highest temperature in spring and winter. Therefore, according to the results presented in Table 6, BSF cannot survive in the spring season.

BSF can quickly degrade organic waste more than municipal waste. This is because the values of different organic and municipal waste parameters are different, and the organic waste parameters are under the faster degradation rate by BSF. On the other side, the suitable direction for the survival of BSF in the North, South, East, and West direction of Pakistan, Southern areas are appropriate for the survival of Black Solider Fly due to its temperature and humidity in summer, winter, and autumn season. BSF cannot survive in harsh conditions such as very low temperatures in Northern areas. Hence, northern areas are not an option for BSF in any season. In the spring season, western areas of Pakistan are suitable for BSF production and growth due to their temperature, humidity, sunlight hours, and precipitation rate. The Eastern areas of Pakistan are not a good place for BSF to live in any season as the temperature there is very high in summer and very low in winter and other seasons of the year, other areas of Pakistan are better for BSF than western areas. Hence, BSF can easily be used to degrade waste in the Southern and Western areas of Pakistan but not for the Eastern and Northern areas of Pakistan.

BSF larvae have demonstrated many social, economic, and environmental benefits to society. A large amount of organic waste reduction is the primary and sustainable means of solid waste management. Diener et al. (2011a) reported a 65%–78% total waste reduction in their research results, depending on the amount of waste added daily. In India's climate conditions, feed on kitchen waste, Pathak et al. (2015) also reported that the larvae could reduce waste by 50% of the original volume. As the larvae greedily eat wastes rich in organic matter, the weight and volume of waste are significantly reduced in a short period. BSF larval technology is economically feasible because the species is self-harvesting and requires less technical skills. The larvae can feed on various organic wastes. According to reports, they feed on straw, distiller's grains, and animal offal (including food waste, fecal sludge, and kitchen waste). Diener et al. (2009) pointed out that BSF larvae are very effective in reducing large amounts of organic waste and at the same time converting waste into protein-rich biomass, which can greatly promote sustainable agriculture.

Another exciting feature of BSF larvae is that they break down organic waste and subsequently produce nutrient residues that can be used as fertilizers. This can be applied to agricultural activities, just like composting or biogas production, which can subtly contribute to the country's revenue. A group of authors found that flies that feed on organic waste can promote the biotransformation of 30 tons of waste per day while producing 33.3% of residues that can be used as fertilizers and 7.7% of pre-pupa biomass can be used for animal feeding. Several authors reported that the manure of larva has analogous qualities to fertilizers and can be used as composed for crop cultivation (Diener et al., 2011a; Choi et al., 2009; Green and Popa, 2012).

For organic waste treatment, bioconversion employing BSF has become a prominent innovation because of its advantages, such as high production rates and cheap costs. The BSF's natural habitat in the wild is to feed on decaying organic materials. There are several reasons for mass production, including the species' capacity to eat a wide range of trash and produce high-quality products under artificial conditions. BSF rearing systems may be adapted to suit the project's specific needs. As a result, when mass production is planned for a certain geographic region, many production factors must be addressed and any unique regulatory criteria of a particular nation. Socio-cultural parameters influence the availability of organic waste, environmental issues, and a viable recycling system. In addition, the greatest economic feature of BSF technology is the capacity of BSFL to transform organic materials into valuable end-products while also needing minimal transport and maintenance expenditures.

Although the insect farming industry has lately grown significantly, several factors must be addressed before this technology can be widely used. These are upgrading BSFL technology. Popularization of BSFL innovation for high-end products. Challenges about manufacturing, shipping, and environmental warehousing effect of BSFL manufacturing processes. Enhancing BSFL-derived item quality. Acceptance of insect-based foods as dietary supplements and education. Aid to low-income or emerging areas to boost BSFL cultivation. Due to the waste being utilized as a substrate throughout the process, the quantity of trash created during food waste treatment with BSFL is decreased. When BSFL absorbs substrates for growth, the substrates are converted into nutritional components that the BSFL may utilize to generate energy. It is feasible to utilize the larval biomass generated by this kind of bioconversion method to develop waste valorization applications further. Because the development of high-quality BSFL biomass is a prerequisite for producing high-quality animal feed and biodiesel, BSFL feeding, and growth are crucial.

## 5. Conclusion and future perspectives

Understanding the physical and chemical composition of MSW is very important for future MSW planning and management. The sustainable management of municipal solid waste (MSW) is one of the main challenges that municipalities face in developing countries such as Pakistan. This study investigated the interactions between municipal and organic waste with BSFL as a growing medium for substrate culture. The study results are based on implementing BSFL technology in terms of municipal solid waste among the four major cities of Pakistan, i.e., Gilgit, Lahore, Peshawar, and Karachi. Our study is the first to provide such a comparison among the major cities of Pakistan. Based on the current investigation, we have evaluated the ability of BSFL on the bioconversion organic waste and municipal waste through the ultimate processing residue under the performance of physicochemical properties of both types of waste. In addition, it can be estimated that there is a significant difference between organic waste and municipal waste. These results further showed that the BSFL could improve the quality of the end product and promote organic waste degradation. Whether organic or solid waste, any type of management requires hours of sustainable and environmentally sound management options.

Many biological methods are in practice to achieve high waste and biomass conversion rates and create value-added chains by synthesizing biomass products. In this case, BSF larval composting is better than the current conventional method. It is recommended that such studies should be conducted in other cities in Pakistan. A comprehensive legal framework regarding integrated MSW management systems should be introduced and implemented in Pakistan. Local municipal authorities should collect the waste collection fee, not by so-called "unions". Finally, low- and middle-income countries must adopt more complex and comprehensive waste treatment methods and renovate them through technological upgrades.

## CRediT authorship contribution statement

**Muhammad Salam:** Data collection, Carried out verification, Tabulation, Statistical analysis of the data. **Amina Shahzadi:** Data collection, Carried out verification, Tabulation, Statistical analysis of the data, Editing and designing of the research. **Huaili Zheng:** Editing and designing of the research. **Fakhri Alam:** Editing and designing of the research. **Ghulam Nabi:** Investigation of manuscript. **Shi Dezhi:** Editing and designing of the research. **Waheed Ullah:** Preparation of the final draft of the manuscript. **Sumbal Ammara:** Preparation of the final draft of the manuscript. **Nisar Ali:** Editing and designing of the research. **Muhammad Bilal:** Preparation of the final draft of the manuscript.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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